# SAAD ZAIDI

*AI & Software Engineer | BS Computer Science — FAST NUCES*

saadzaidi129@gmail.com • 0330-2740209 • github.com/SaadZaidi1 • linkedin.com/in/saad-zaidi-42a253357 • saadzaidi1.github.io

## PROFESSIONAL SUMMARY

Dean's List student at FAST NUCES with hands-on experience building intelligent ML systems and production-grade server-side services. Proven ability to design deep learning pipelines — from raw sensor data to deployable models — using PyTorch, scikit-learn, and Hugging Face, while delivering scalable RESTful endpoints, microservices architectures, and containerized deployments. Adept at shipping complete, end-to-end solutions that are measurably impactful, technically rigorous, and built for scale.

## EDUCATION

**FAST National University of Computer and Emerging Sciences** *2023 – 2027*

*Bachelor of Science in Computer Science* **| Dean's List – 3rd Semester**

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, C/C++, Assembly
**Server-Side Frameworks:** Flask, FastAPI, RESTful Web Services, JWT Authentication, Celery, Redis
**ML / AI:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face Transformers, OpenAI API, SHAP, Streamlit
**Databases:** PostgreSQL, SQLite3, MySQL
**DevOps & Tools:** Docker, Kubernetes, Railway, GitHub Actions, Render, Linux, Git/GitHub, Jupyter, Pandas, NumPy
**Frontend:** HTML, CSS, JavaScript, React
**CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming

## PROJECTS

**Intelligent CNC Process Monitoring & Defect Prediction** | *PyTorch, XGBoost, scikit-learn, SHAP, Pandas*

- Developed a 3-layer ML pipeline: LSTM Autoencoder for anomaly detection (F1 = 0.9316, +43.3% over baseline), 1D-CNN + XGBoost for tool-wear classification (87.95% accuracy, F1 = 87.86%), and stacked BiLSTM + Bahdanau Attention for RUL prediction ($R^2$ = 0.5930, +21.9%, MAE = 0.1343 µm).
- Extracted 1,000+ FFT, statistical, and mechanical features from 18 GB of sensor data across three CNC datasets; applied SMOTE oversampling for class balancing and SHAP TreeExplainer for full feature-level explainability.

**Crowd-Funding Platform** | *Flask, FastAPI, PostgreSQL, Docker, JavaScript* *Backend Lead*

- Architected a production-grade RESTful service layer with ACID-compliant transactions, role-based access control, and input validation — sustaining 100+ concurrent campaign operations with zero data integrity failures.
- Designed a RAG-based FAQ microservice using vector embeddings in FastAPI, containerized with Docker; reduced inbound support query volume by 40% through intelligent automated resolution.
- Implemented microservices architecture decoupling the RAG service from the core application, enabling independent horizontal scaling of each component.

**RO-Plant Customer Portal** | *Flask, JavaScript, SQLite3* *Freelance · 2025*

- Delivered a freelance operations dashboard tracking daily bottle sales, customer accounts, and live profit totals — eliminating paper-based reconciliation and reducing daily close-out time by 96% (2 hrs → 5 min).

**LLM Engineering** | *Python, Jupyter, Hugging Face, OpenAI API*

- Conducted prompt-engineering experiments across GPT-4 and open-source LLMs, achieving 30%+ improvement in task-specific accuracy; delivered modular production-ready applications for text generation, summarisation, and classification.

## CERTIFICATIONS

**LLM Engineering** — Udemy (Ed Donner) · 2024